Car Rider Pickup System for Schools

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The Wranglers

Problem Background

- Over 70% of students are now picked up via car, creating massive congestion.
- Parents wait up to 1 hour, with lines often forming 1-2 hours before dismissal.
- Queues spill onto local highways, causing traffic backups and increasing accident risks.
- Inclement weather increases delays as bus and bike riders often switch to the car line.
- The manual process requires 5-10 staff members daily, diverting them from educational duties.

Needs Statement

School pickup via the car rider line has been increasing in the US over the years leading to an increase in line length and pickup time. Now, over 70% of students are picked up from school via the car rider line. Car rider lines have become congested, posing a safety issue for both the students and other drivers. With more students attending school, car rider lines have become too slow and cause parents to wait up to over an hour in line leading to frustration. Some studies show that the car rider pick-up line starts to form 1-2 hours before school actually gets out, which leads to traffic backing up local highways and streets, and even frequent accidents. Inclement weather further slows the line, as students who ride the bus or bike to school will often be picked up by their parents in the car rider line to avoid the weather. Additionally, 5-10 staff members are required to set up and run the pickup process, diverting their time away from the education of students. An automated system is needed to streamline the pick-up process to decrease delays and safety issues.

Goal and Objectives

Goal

The goal of this project is to design and prototype a system that will help automate the car rider pick up process at schools. The system is built to reduce the amount of time necessary to pick up students from school.

Objectives

- The system will allow parents to update their information (license plate numbers, approved adults for pickup, car rider schedules) via a website with approval from a staff member of the school.
- The system will read the marker on the car (license plate, QR code, nametag) as it enters the pickup line and send the information to a server.
- The server will identify the parent and child associated with the car marker, and update a display at the entrance of the pickup line to direct the car to the proper station.
- The laptop at the front of the school is updated to inform the students and staff which child is being picked up, and which zone to go to.

Goal and Objectives - Requirements

Marketing Requirements

1. Make the pickup process faster for parents
2. The system should be easily updated
3. The system should be low cost
4. The system should be easy to use/install
5. Information must be validated if updated or inserted
6. Student safety should be ensured (information, physical, etc.)
7. The system should be reliable

Goal and Objectives - Requirements

Marketing Requirements	Engineering Requirements
1	Pickup time should be reduced by 30% per 100 kids. The time from the car entering the view of the camera to information being displayed should be no more than 5 seconds.
4	7 inch (diagonal) display screen to give station and process instructions clearly.
1, 6	Per Texas Family Code, ensure that students are referred to authorized parents 100% of the time.
6	Private Information stored in the database should be encrypted using 128 bit encryption.
7	Should have a stable connection from within 1 kilometer, stable power to all components.
1,6,7	Should correctly identify/assign parents to kids 100% of the time.
3	The costs for developing the system should not exceed $300.
2,4,5	The system should allow parents to submit a request to the school to update information and the school should receive the request all within 1 minute.

Goal and Objectives - Requirements

Marketing Requirements	Engineering Requirements
4	The system should have a modular design such that failed components can be replaced by a technician in under 15 minutes.
4,7	The system should be mounted correctly on a tripod to ensure stability and prevent hazards. Sandbag/Weights of at least 1 pound to keep it grounded.
7	The software will comply with the Digital Millennium Copyright Act.
2,4,5,6	Staff should be able to access databases with provided credentials and view all parent to children links.
4,7	The system should be able to withstand 20mph wind.
1	Minimum 40inch (diagonal) display screen to give station and process instructions clearly inside in the pickup .
4	The system should have a modular design such that failed components can be replaced by a technician in under 15 minutes.
4,7	The system should be mounted correctly on a tripod to ensure stability and prevent hazards. Sandbag/Weights of at least 1 pound to keep it grounded.

Literature Review -

Overview

- Schools use technology to reduce pickup congestion, improve safety, and streamline communication
- These systems inform our design, though none use our proposed method

Existing Systems

- **PikMyKid** – App-based digital car line; parents announce arrival, staff release students via dashboard
- **Beeline** – Uses geofencing to detect vehicles and alert staff automatically
- **Car-Pool App** – Geofencing and ETA alerts to stage students before arrival
- **iSchoolRide** – GPS-based parent check-in for just-in-time student release
- **Kids Pick Up App** – QR code scanning to verify authorized pickups

Key Takeaway

- Current solutions reduce manual effort and congestion
- Most still rely on parent apps, GPS, or staff scanning and verification

Design Constraints and Feasibility

Technical and Temporal Constraints

- Maximum of 5 seconds from image capture to display update
- Identification accuracy must be 100% to ensure safety and efficiency
- Connection between the system parts must be reliable

Physical and Environmental

- Needs to withstand outdoor elements (wind and rain)
- Needs to be portable and stable with modular parts, mounting, and power supply to reduce system downtime and speed up repair and maintenance

Economic

- Must cost $250 or less, necessitating cost-effective components

Feasibility

- Core technology, such as Python image processing and Wi-Fi, is well documented and achievable
- Main challenge will be creating a fast and efficient system within our budget

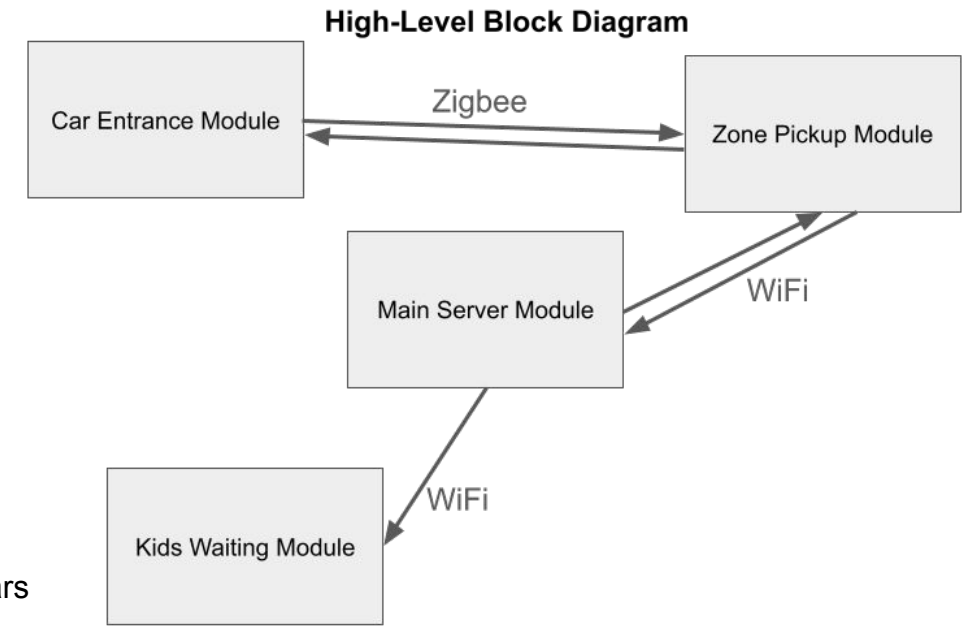
Evaluation of alternative solutions

Solution	Pros	Cons
Manual Pickup App (PikMyKid)	Easy to implement; low cost; reduces paperwork	Requires staff queue management; no automated vehicle identification
Integrated Dismissal System (Beeline)	Improves communication; integrates with school systems	Requires training and infrastructure; still staff-monitored
Geofencing (Car-Pool App)	Real-time arrival alerts; reduced manual check-ins	Dependent on GPS accuracy, mobile devices, and connectivity
GPS Check-In (IschoolRide)	Keeps students inside until arrival; reduces idle staging	Relies on parent app usage and accurate GPS; staff verification will still be needed
QR Codes (Kids Pick Up App)	Simple, affordable; no cellular service required	Requires staff scanning; affected by glare, weather, and visibility

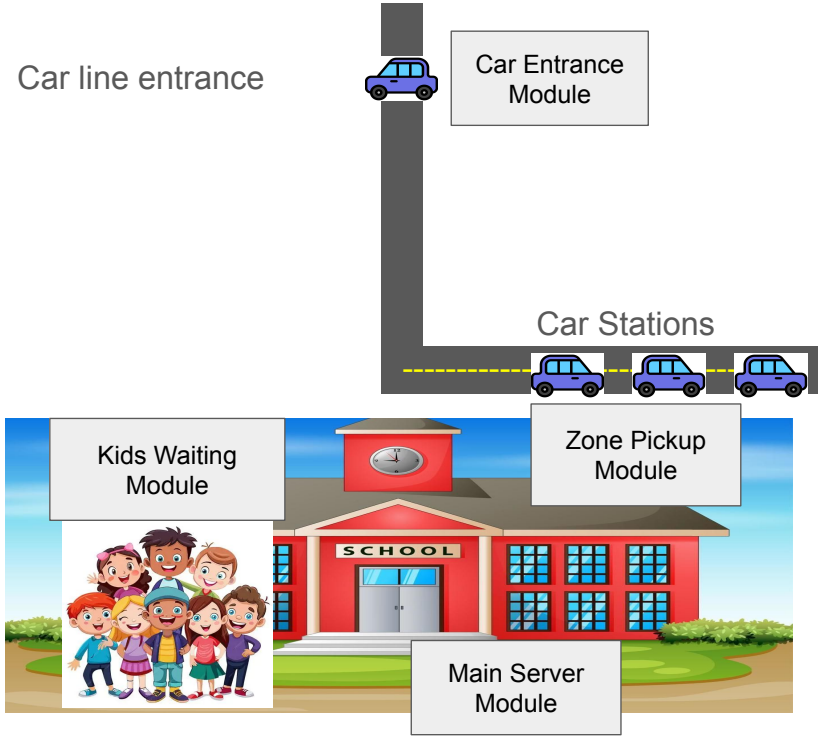
All alternatives improve organization but still rely on staff actions or parent interaction, limiting full automation.

Proposed design - High-Level Block Diagram

- Need 4 main modules
  - Car Entrance Module
    - Located at entrance of line
  - Zone Pickup Module
    - Located at pickup zones
  - Main Server Module
    - Located inside school's office
  - Kids Waiting Module
    - Located at area where kids wait to be called (gym, cafeteria, outside)
- Zigbee - low power wireless mesh network standard 0-100m (similar to Bluetooth)
  - More reliable than Bluetooth for small packets of data
  - Avoid Bluetooth interference from passing cars
  - No stable WiFi at entrance



Proposed design - High-Level Block Diagram



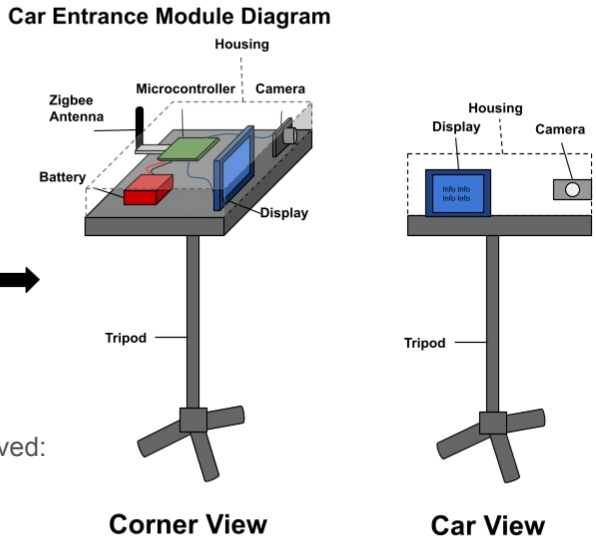
Proposed design - Car Entrance Module

- Primary function is to complete license plate recognition
  - Camera takes frequent pictures
  - Processes image for license plate
  - Once found, send license plate number using Zigbee
- Display for parents
  - Wait for results from server, then update display to confirm correct kid and direct parent to specific zone
- 3D printed housing
  - Protect components from direct sunlight and light rain

Example packet of info sent:


```
{
 "event": "license_plate_detected",
 "plate": {
   "number": "ABC1234",
   "state": "TX"
 }
}
```


Car Entrance Module



Example packet of info received:

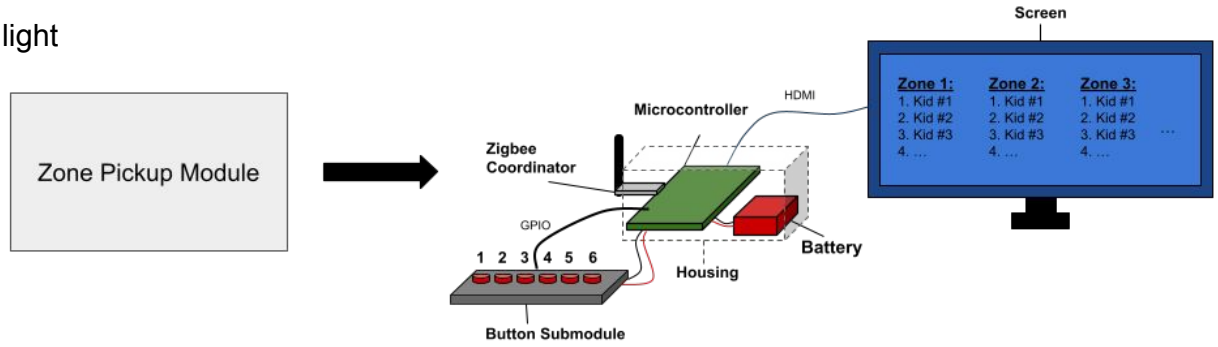

```
{
 "event": "update_display",
 "info": {
   "name": "John Doe",
   "zone": "1, red"
 }
}
```

Proposed design - Zone Pickup Module

- Act as a receiver for Zigbee communication - forward info to server (WiFi)
  - Also forward info from server to car entrance module (Zigbee)
- Large screen to display website at the zones - lists of kids being picked up at each zone
- Button submodule - allows staff member to press a button when a kid is picked up at a zone, one button for each zone
  - Removes kids name from list

- 3D printed housing
  - Block direct sunlight and light rain

Zone Pickup Module Diagram



Proposed design - Main Server Module

- Located in school's office
- Host a website - updated in real time
  - Displayed on screens in zone pickup and kids waiting modules
- Connect to database that contains info about all kids parents, pickup schedule, and license plate numbers.
  - Query the database to find info given license plate number
  - Update database based on requests made by parents (once approved by admin)
- Send updates to display in car entrance module
  - After querying database

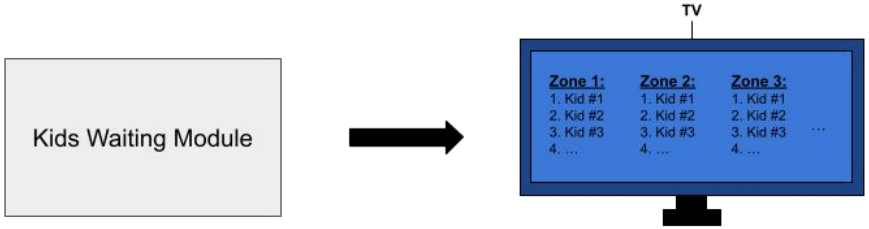
Main Server Module Diagram

Server ⟷ WiFi ⟷ Zone Pickup Module ⟷ Zigbee ⟷ Car Entrance Module

Proposed design - Kids Waiting Module

- Large TV screen where kids are gathered to wait (gym, cafeteria, outside)
  - Display website to inform staff and kids who is being picked up, and what zone to go to

Kids Waiting Module Diagram



Approach for design validation

Small, foundational testing first

- Verify connectivity to web server and database
- Display text found by the camera directly to the screen it is connected to
- Ping router from the car entrance module to verify connectivity
- QUERY database to ensure proper initialization

System testing comes next

- Packets should be sent to the database, should return student names
- Screens directing drivers and students should update in no more than 5 seconds
- Web server flags incoming requests to alter database to admin page

Economic analysis and budget

Economic Viability:

- National Center for Education: There are 70,055 Public Pre-Kindergarten, Elementary and Middle Schools
  - Each school has to have a pickup system
  - Could also be used in Daycares and Private Schools
- Save Parents & Teachers Time
  - Allow them to spend time in more productive ways
- Production costs for items and the 3D casing would decrease if ordered and produced in bulk

Economic analysis and budget

Sustainability:

- Raspberry Pi, Camera and Zigbee are all available from multiple vendors
- Standard software frameworks are used to ensure long-term support
- Limited Customer maintenance:
  - Charging of rechargeable batteries
  - Yearly website database setup
  - Software updates

Economic analysis and budget

Manufacturability:

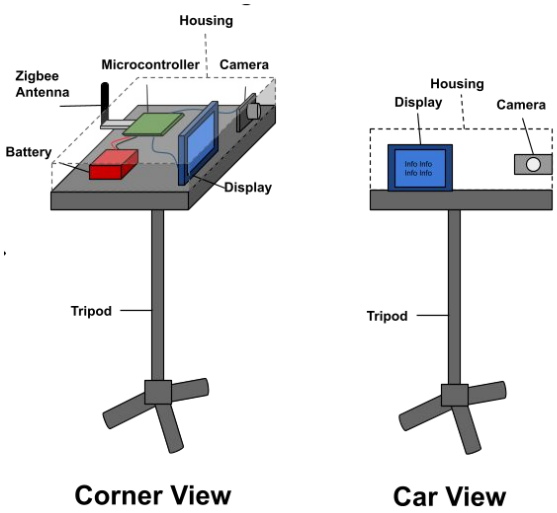
- Camera alignment/height needs to be correct to ensure licence plate can be read
- Casing of Camera designed to withstand wind and rain
- Modular Design allows for easy replacement of broken components
- In the event that the license plate doesn’t match, we have a separate spot for pickup
- End-to-End is testable as well as component based testing
- Compliant with student privacy regulations

Economic analysis and budget

- Car Entrance Module

Car Entrance Module	Amount Needed	Cost per Unit	Total Amount
Tripod	1	$22	$22
Raspberry Pi	1	$75	$75*
Raspberry Pi Camera	1	$55	$55*
Camera Lens	1	$50	$50*
Zigbee Antenna	1	$27	$27
Rechargeable Battery	1	$40	$40
7-inch display	1	$38	$38*
Mounting Screws	~20	$8 (600 pcs)	$8
Wires	~5	$10 (240 pcs)	$10
Total* (without including components provided)	**$107**		
Total Cost	**$325**		

- **Total* (without including components provided): $107**
- **Total: $325**

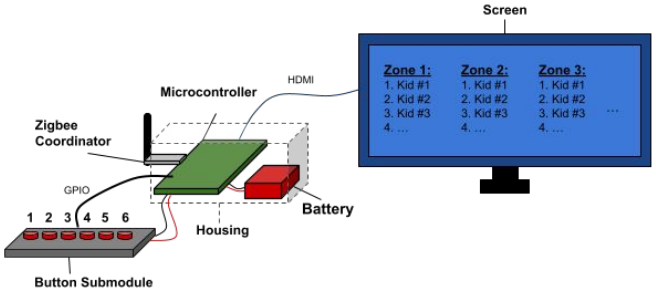


Economic analysis and budget

- Zone Pickup Module

Zone Pickup Module	Amount Needed	Cost per Unit	Total Amount
Raspberry Pi	1	$75	$75*
Zigbee Antenna	1	$27	$27
Rechargeable Battery	1	$40	$40
27-inch monitor	1	$97	$97*
Push Buttons	6 (depending on # of zones)	$11.50 (24 pcs)	$11.50
Mounting Screws	~20	$8 (600 pcs)	$8*
Wires	~15	$10 (240 pcs)	$10*
HDMI Cord	1	$4	$4
Total* (without including components provided)	**$82.50**		
Total Cost	**$272.50**		

- **Total* (without including components provided): $82.50**
- **Total: $272.50**



Economic analysis and budget

- Main Server Module
- Assuming server is hosted on a computer used by the school

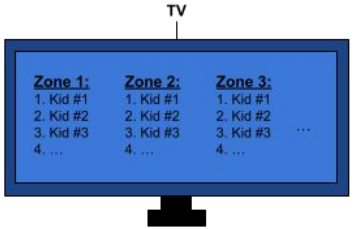
Main Server Module	Amount Needed	Cost per Unit	Total Amount
Total Cost	**$0**		

- **Total: $0**

- Kids Waiting Module

Main Server Module	Amount Needed	Cost per Unit	Total Amount
55-inch TV	1	$180	$180*
Total* (without including components provided)	**$0**		
Total Cost	**$180**		

- **Total* (without including components provided): $0**
- **Total: $180**



Economic analysis and budget

- Total budget without any components provided:

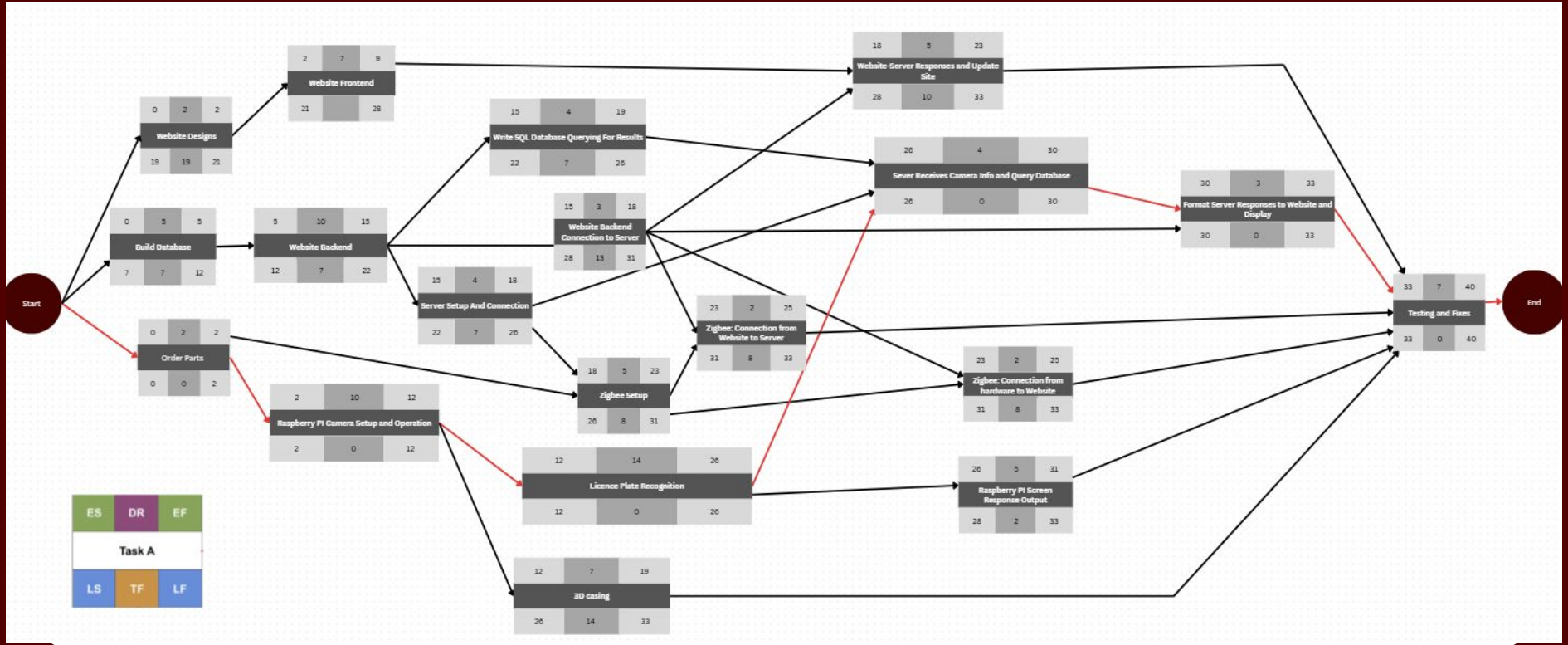
Car Entrance Module	$325
Zone Pickup Module	$272.50
Main Server Module	$0
Kids Waiting Module	$180
Total	**$777.50**

- Total budget with provided components:

Car Entrance Module	$107*
Zone Pickup Module	$82.50*
Main Server Module	$0
Kids Waiting Module	$0*
Total*	**$189.50**

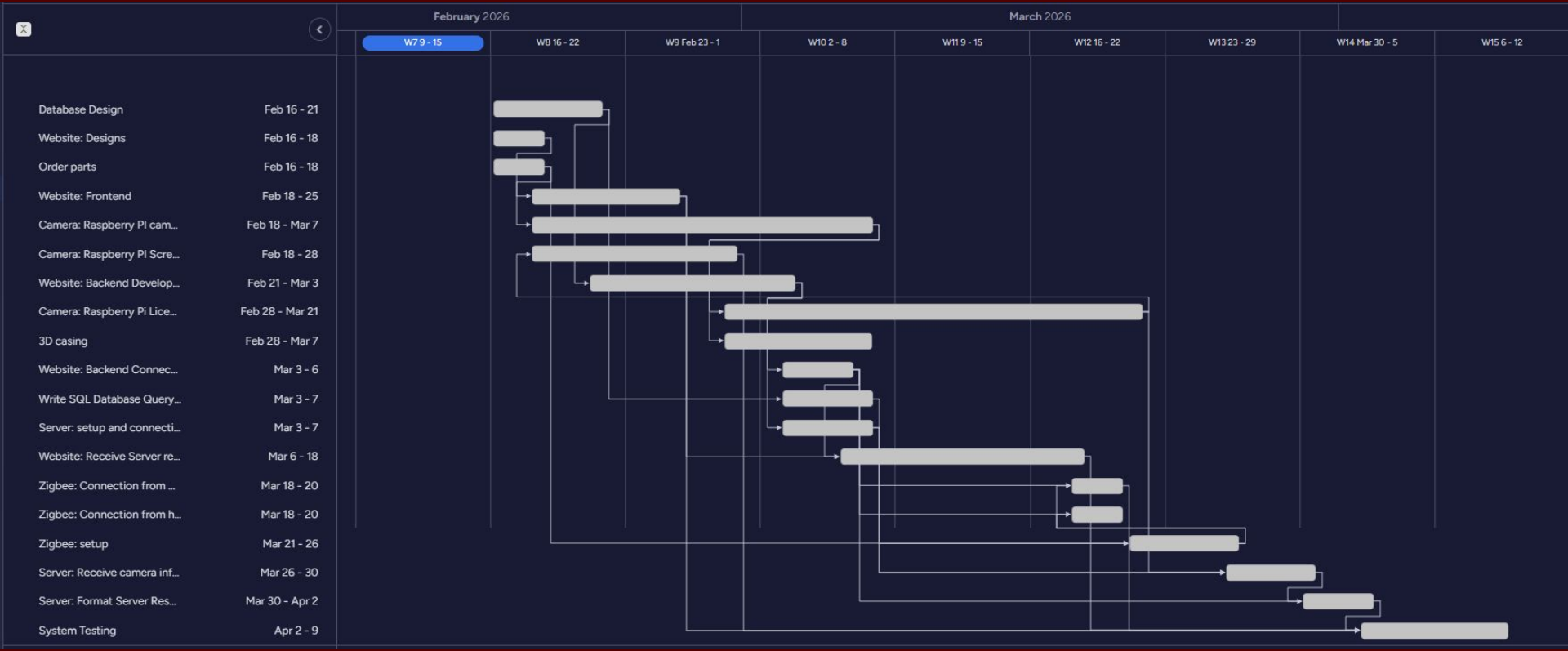
- Note*: 3D printed prototypes not included, FEDC grants $75 allotment for fabrication

Schedule of tasks - Pert Chart



Critical Path: 45 days

Schedule of tasks - Gantt Chart



Schedule of tasks

Name	Start Date	End date	Time	Task description:
Order parts	**2/16/2026**	**2/18/2026**	2	Order all remaining parts
Website: Designs	2/16/2026	2/18/2026	2	Create wireframes for Parent Portal Page, Login Screen, Admin Dashboard, and Kids Pickup Status Screen, and get group approval of designs
Build Database	2/16/2026	2/21/2026	5	Create database schema, basic CRUD queries and set up authentication-related database endpoints
Website: Frontend	2/18/2026	2/25/2026	7	Implement approved designs with responsive layouts
Camera: Raspberry PI camera setup and operation	2/18/2026	2/28/2026	10	Install Raspberry Pi OS, enable camera interface, install required libraries and test image capture functionality
Website: Backend Development	2/21/2026	3/3/2026	10	Set up backend framework, connect it to the database, implement authentication and authorization logic, validate user input, and implement basic logging

Schedule of tasks

Name	Start Date	End date	Time	Task description:
Camera: Raspberry Pi Licence Recognition	2/28/2026	3/21/2026	14	Develop license plate recognition system using Raspberry Pi camera
3D casing	2/28/2026	3/7/2026	7	Design casing for Raspberry Pi camera system
Server: setup and connection	3/3/2026	3/7/2026	4	Configure server environment and set environment variables, and ensure secure database connectivity
Write SQL Database Querying For Results	3/3/2026	3/7/2026	4	Write SQL Queries for pickup to match license plate numbers to authorized parents
Website: Backend Connection to Server	3/3/2026	3/6/2026	3	Deploy backend application to server environment
Zigbee: setup	3/6/2026	3/18/2026	5	Install Zigbee coordinator and modules

Schedule of tasks

Name	Start Date	End date	Time	Task description:
Zigbee: Connection from hardware to the Website	3/18/2026	3/20/2026	2	Configure server to update Website when hardware sends plate
Camera: Raspberry PI Screen Response Output	3/21/2026	3/26/2026	5	Make the screen show green when recognized license plate number is verified & handle error or unrecognized plate cases
Server: Receive camera info and query the database	3/26/2026	3/30/2026	4	Implement server-side endpoint to receive license plate data from Raspberry Pi, validate input, and query database for matching parent/student
Server: Format Server Responses to Website and Display	3/30/2026	4/2/2026	3	Send verification results to both classroom display system and website frontend
System testing	4/2/2026	4/9/2026	7	Test end-to-end functionality of entire system

Project management and teamwork

Haya

- Hardware Design
- OCR Design

Jake

- Web Applications
- OCR Design

Jack

- Team Leader
- Systems Design

Jayden

- Hardware Design
- Systems Design

Sara

- Database Design
- Web applications

Everyone

- Software Engineering
- Documentation/ Reports
- Testing

monday.com will be used for project management. It will track dependencies and division of labor, as well as provide a timeline for the project

Societal, safety and environmental analysis

Societal:

- Parents spend less time waiting in line
- Teachers spend less time managing pickup and more in classroom
- Kids are less likely to get agitated/distracted leading to less problems for staff/parents
- The secure system ensures that students are released to the correct parents

Safety:

- Ensure that kids end up with the correct parent
- No kid is lost in the system or placed incorrectly
- Ensure that kids aren’t put in dangerous situations (ex running in the road)

Environmental:

- Reduction in vehicle idling

Questions?

Thanks for listening!

The Wranglers